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| المادة : Ordinary Differential Equations | جامعة طبرق - كلية العلوم | امتحان: الدور الأول |
| اليوم : 2026 / 01 / 31 م | University of Tobruk 1990 | القسم : الفيزياء |
| الزمن : ساعتان | | المرحلة : الثانية (الفصل الثالث) |

**Q1: a) Find the equation from the general solution:**

1) $y = c\,e^{2x}$
2) $y = c_1 e^{-x} + c_2 e^{-3x}$

**b) prove that the function $y = e^{2x}$ is solution to the equation:**

$$y'' - 4y = 0$$

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**Q2: By separation of variable find the general solution.**

$$4y\,dx - x(y-3)\,dy = 0$$

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**Q3: Solve the homogeneous equation:**

$$x\,dy - (2y + x)dx = 0$$

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**Q4: Solve the non-exact equation:**

$$3x^3y^2\,dx + (2x^4y + 4x\,y^3)dy = 0$$

**Q5:- Solve the linear equation**

$$\frac{dy}{dx} + \frac{1}{x}y = x^3 - 3$$

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**Q6: Find the solution of the differential equation (Bernoulli's):**

$$\frac{dy}{dx} - \frac{y}{2x} = 5x^2y^5$$

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مع تمنياتنا بالتوفيق والنجاح ...

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| امتحان: الجزئي<br>القسم : الرياضيات + الفيزياء<br>المرحلة : الثانية (الفصل الثالث) | جامعة طبرق - كلية العلوم<br>1990<br>University of Tobruk | المادة : Ordinary Differential Eqyations1<br>اليوم : 29 / 11 / 2025 م<br>الزمن : ساعتان |

**Q1: a)Find the equation from the general solution:**

1) $y = c \sin x$

2) $y = c_1 e^{2x} + c_2 e^{3x}$

**b) Prove that the function $y = 3e^{-x}$ is solution to the equation:**

$$y' + y = 0$$

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**Q2: By separation of variable find the general solution.**

$$\frac{dy}{dx} = \frac{x+1}{2-y}$$

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**Q3: Prove that the differential equation is exact then find the general solution.**

$$(2x^3 - xy^2 - 2y + 3)dx - (x^2y + 2x)dy = 0$$

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**Q4: Solve the non-exact equation.**

$$y\,dx - x\,dy = 0$$

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